

Cache Tier Manager

In database caching, managing data efficiently between fast RAM and slow disk storage is critical for system performance. A two-tier storage system classifies cached data as either **HOT** (frequently accessed, fast retrieval) or **COLD** (infrequently accessed, slower retrieval) to optimize resource usage and query latency.

Your task is to implement a class `TopKCache`` that manages data movement between two storage tiers. The system keeps all data but classifies keys into **HOT** (top K most recently used) or **COLD** (all other stored keys) based on usage recency. The cache must support insertions, retrievals, and status queries while maintaining correct tier classification.

Class Methods

- `__init__(self, k)``: Initializes the cache with a **HOT** capacity of `k`` key-value pairs.
- `put(self, key, value)``: Stores the key-value pair in the cache and marks it as the most recently used.
- `get(self, key)``: Returns the value corresponding to `key`` and marks it as the most recently used. Returns `None`` if the key is not present.
- `get_status(self, key)``: Returns `HOT`` if the key is in the top K recently used keys, `COLD`` if it exists but is not in the top K, or `None`` if missing. This method **does not modify the recency order**.

Example

Input:

```
8
init 2
put A 10
put B 20
put C 30
status A
get A
status A
status B
```

Output:

```
COLD
10
HOT
COLD
```

Explanation:

- `init 2`: Top 2 items are HOT.
- `put A, put B`: The order is [B, A]. Both are HOT.
- `put C`: Order is [C, B, A]. The recent 2 keys, C and B are HOT, while A becomes COLD.
- `status A`: Returns the string "COLD".
- `get A`: A is accessed. The order becomes [A, C, B]. The recent 2 keys, A and C are HOT, while B becomes COLD.
- `status A`: Returns the string "HOT".

- status B: Returns the string "COLD".

Note:

Constraints

- $1 \leq \text{Number of operations (Q)} \leq 10^5$.
- $1 \leq K \leq 1000$
- keys and values are alphanumeric strings.

Input Format for Custom Testing

Input from stdin will be processed as follows and passed to the class methods:

The first line contains an integer Q, the total number of commands. Each of the next Q lines contains a command in one of the following formats:

- ``init <k>``: Initializes TopKCache with capacity k
- ``put <key> <value>``: Calls ``put(key, value)``
- ``get <key>``: Calls ``get(key)`` and prints the result
- ``status <key>``: Calls ``get_status(key)`` and prints the result